# MATH 110 — Week 4

## Homework 4 — Solutions

**HW4 P1.**

**Idea.** Eliminating one unknown in the two output equations gives the coefficients of the proposed inverse. When the determinant is zero, a nonzero kernel vector or a missing output rules out invertibility.

1. Let $B = \left(\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right)$. Multiplication gives $AB = \left(\begin{smallmatrix} ad-bc & -ab+ba \\ cd-dc & -bc+ad \end{smallmatrix}\right) = (ad-bc)I$, and likewise $BA = (ad-bc)I$.
2. If $ad - bc \neq 0$, then $A((ad-bc)^{-1}B) = ((ad-bc)^{-1}B)A = I$, giving the stated inverse formula.
3. If $ad - bc = 0$ and $(a, b) \neq (0, 0)$, then $(b, -a) \neq 0$ and $A(b, -a) = (0, cb - da) = 0$; thus $A$ is not injective.
4. If $a = b = 0$, every output has first coordinate zero, so $A$ is not onto. Either obstruction excludes invertibility by Proposition 29.

**HW4 P2.**

**Idea.** Use the two-by-two determinant in the correct field.

1. Over the given field $F$, $2 \cdot 3 - 1 \cdot 5 = 1$, so the inverse is $\left(\begin{smallmatrix} 3 & -1 \\ -5 & 2 \end{smallmatrix}\right)$. Direct products with the original matrix are $I_2$.
2. Over $\mathbb{C}$, $1(-1) - i^2 = -1 + 1 = 0$, so the matrix is singular (not invertible). The nonzero kernel vector $(i, -1)$ confirms it.
3. Over $\mathbb{F}_5$, $1 - 6 = -5 = 0$, so the matrix is singular. The nonzero vector $(3, 1)$ has image $(5, 10) = (0, 0)$ modulo 5.

**HW4 P3.**

**Idea.** An isomorphism preserves basis spanning and independence; conversely one image basis already makes the map onto.

1. Assume $T$ is an isomorphism and let $v_1, \ldots, v_n$ be any basis. A relation $\sum a_j T v_j = 0$ implies $\sum a_j v_j = 0$ by injectivity, so all coefficients vanish.
2. Every $w$ equals $Tv$ by surjectivity, and expanding $v$ in the basis expresses $w$ in the image list. Thus that list is a basis.
3. Conversely choose a basis of finite-dimensional $V$; the hypothesis says its image list spans $V$, so $T$ is onto. Proposition 31 gives invertibility. This includes $V = \{0\}$ and its empty basis.

**HW4 P4.**

**Idea.** Every BC output is a B output; also BC only uses inputs to B lying in range C. These give the two bounds.

1. Every vector $BCx$ lies in $\operatorname{range} B$, so $\operatorname{rank}(BC) \leq \operatorname{rank} B$.
2. The restriction $B|_{\operatorname{range} C}$ has range $\operatorname{range}(BC)$. Rank–nullity on its domain gives $\dim \operatorname{range}(BC) \leq \dim \operatorname{range} C = \operatorname{rank} C$. Combining yields the minimum bound.
3. For $B = \operatorname{diag}(1, 0)$ and $C = \operatorname{diag}(0, 1)$, both ranks are 1 but $BC = 0$ has rank 0.

### HW4 P5.

**Idea.** Put each factor basis into its own coordinate of the product.

1. If $v_1, \dots, v_r$ and $w_1, \dots, w_s$ are factor bases, use $(v_1, 0), \dots, (v_r, 0), (0, w_1), \dots, (0, w_s)$.
2. Each $(v, w)$ expands using the separate factor expansions, proving spanning. A zero relation gives $\sum a_i v_i = 0$ and $\sum b_j w_j = 0$, so all coefficients vanish by independence.
3. The list is a basis of length $r + s = \dim V + \dim W$.

### HW4 P6(a).

**Idea.** Any subspace must contain zero, which determines when a translate is the original subspace.

1. If $v + U$ is a subspace, $0 = v + u$ for some $u \in U$, so $v = -u \in U$.
2. If $v \in U$, then $v + u \in U$ for all $u \in U$, giving $v + U \subseteq U$. Conversely $u = v + (u - v) \in v + U$ for every $u \in U$. Hence $v + U = U$, a subspace.

### HW4 P6(b).

**Idea.** Equality of translates is equivalent to their representative difference lying in the subspace.

1. If $v + U = v' + U$, then $v = v' + u$ for some $u \in U$, so $v - v' \in U$.
2. If $v - v' \in U$, each $v + u = v' + (v - v' + u)$ lies in $v' + U$. Reversing the roles gives equality, as in Proposition 39.

### HW4 P7 (optional).

**Idea.** The missing linearity hypothesis is essential; with it, two extended bases prescribe the desired isomorphism.

1. As printed, take $U = V = \mathbb{R}$ and $S(x) = x^3$. This is injective but not linear: $S(2) = 8 \neq 2S(1)$. An extension on all of $V = U$ would equal $S$, so no linear extension exists.
2. Assume now that $S$ is linear. Choose a basis $u_1, \dots, u_k$ of $U$ and extend it to $u_1, \dots, u_k, v_{k+1}, \dots, v_n$ of $V$.
3. If $0 = \sum_i a_i S u_i = S(\sum_i a_i u_i)$, injectivity gives $\sum_i a_i u_i = 0$; independence of the $u_i$ forces all $a_i = 0$. Thus $Su_1, \dots, Su_k$ is independent. Extend this image list to a basis $Su_1, \dots, Su_k, w_{k+1}, \dots, w_n$ of $V$.
4. Prescribe $Tu_i = Su_i$ and $Tv_j = w_j$. The rule $T(\sum a_i u_i + \sum b_j v_j) = \sum a_i S u_i + \sum b_j w_j$ defines a unique linear $T$ by the basis rule; it maps a basis to a basis, so P3 makes it invertible. On $U$, expansion in the $u_i$ shows $T = S$.

### Exam practice, Question 2.

**Idea.** A left inverse must send $Tv_j$ back to $v_j$. Injectivity makes those prescribed inputs independent, so a basis extension permits defining the remaining values freely.

1. If $\sum_j a_j T v_j = 0$, linearity gives $T(\sum_j a_j v_j) = 0$. Injectivity gives $\sum_j a_j v_j = 0$ and basis independence gives all $a_j = 0$. This proves (a).
2. Extend $Tv_1, \dots, Tv_n$ to a basis $Tv_1, \dots, Tv_n, w_{n+1}, \dots, w_m$ of $W$ (Week 2 Proposition 11). Define $S(\sum_j a_j T v_j + \sum_{k>n} b_k w_k) = \sum_j a_j v_j$. Unique coordinates make this linear.
3. For $v = \sum_j a_j v_j$, $STv = \sum_j a_j S(Tv_j) = \sum_j a_j v_j = v$. Thus $ST = I_V$, proving (b). A left inverse need not be a right inverse when dimensions differ: the inclusion $t \mapsto (t, 0)$ and $S(x, y) = x$ satisfy $ST = I$ but $TS(x, y) = (x, 0)$.

### Exam practice, Question 3.

**Idea.** Compute basis images, then use the equal-dimension criterion.

1. The three images of $B$ are $1 + x^2$, $2 + x + x^2$, and $2 + 2x + 2x^2$. Thus $M_C^B(T) = \begin{pmatrix} 1 & 2 & 2 \\ 0 & 1 & 2 \\ 1 & 1 & 2 \end{pmatrix}$.
2. If $T(a, b, c) = 0$, coefficient comparison gives $a + b = 0$, $b + c = 0$, $a + c = 0$. The first two give $b = -a, c = a$; the last gives $2a = 0$, hence $a = b = c = 0$ over $\mathbb{R}$.
3. The map has zero kernel between two spaces of dimension 3, so Proposition 31 gives an isomorphism.

**Exam review, Exercise 9.**

**Idea.** The tuple $(u, w)$ supplies the two summands; the zero-intersection assumption makes their sum determine the tuple.

1. Define $\Gamma : U \times W \to U \oplus W$ by $\Gamma(u, w) = u + w$. For a scalar $a$, $\Gamma(a(u, w) + (u', w')) = a(u + w) + (u' + w')$, so $\Gamma$ is linear.
2. It is onto by the definition of the sum. If $\Gamma(u, w) = 0$, then $u = -w \in U \cap W = \{0\}$, so $u = w = 0$. Thus it is injective and Proposition 29 makes it an isomorphism.

**Exam review, Exercise 10.**

**Idea.** The right shift keeps every input coordinate, but constrains the first output coordinate to zero.

1. Its rule is linear coordinate by coordinate. If $Ra = Rb$, comparing coordinate $j + 1$ gives $a_j = b_j$ for every $j \geq 1$, so it is injective.
2. The sequence $(1, 0, 0, \ldots)$ is not an output because every output has first coordinate zero. Thus it is not onto.
3. For an operator on finite-dimensional $V$, injectivity gives $\dim \operatorname{range} T = \dim V$ by rank–nullity, so the range must be $V$. Equivalently Proposition 31 makes injectivity and surjectivity equivalent. Hence this behavior is impossible there.